

Week-4 Module

Q1: Printing integers 1 to N.

```
import java.util.Scanner;
public class demo {
    public static void main (String [] args) {
        Scanner sc = new Scanner (System.in);
        int n = sc.nextInt();
        for (int i=1; i<=n; i++) {
            System.out.print (i+ " ");
        }
        sc.close();
    }
}
```

Explanation : Time Complexity -
loop (for loop) run exactly n times
input n.
time = $n \times O(1) = O(n)$

space complexity - Constant space
Order - $O(1)$

This program prints every inter from 1 to N
in increasing order.

Q2: Sum of first N natural no using loop.

```
import java.util.Scanner;
public class demo {
    public static void main (String [] args) {
```

```

Scanner sc = new Scanner (System.in);
int n = sc.nextInt();
int sum = 0;
for (int i = 1; i <= n; i++) {
    sum = sum + i;
}
System.out.println (sum);
sc.close();
}
}

```

Time Complexity - loop run n times
 input n , fixed time; time complexity $O(n)$
 Space Complexity - Constant space $\Rightarrow O(1)$

Q3. A raised to the power B.

```

import java.util.Scanner;
public class demo {
    public static void main (String [] args) {
        Scanner sc = new Scanner (System.in);
        int a = sc.nextInt();
        int b = sc.nextInt();
        int val = 1;
        for (int i = 1; i <= b; i++) {
            val = val * a;
        }
        System.out.println (val);
        sc.close();
    }
}

```

space complexity $\Rightarrow O(1)$ const. space
 time complexity $\Rightarrow$ Input b ; ~~loop~~ const. $O(1)$
 $\searrow$
 $O(b)$

ew Go Run ... ← → Untitled (Workspace) Update

J demo.java X

.vscode > r coding > J demo.java > demo

```
1 import java.util.Scanner;
2 public class demo {
3     Run | Debug
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int n = sc.nextInt();
7         for (int i = 1; i <= n; i++) {
8             System.out.print(i + " ");
9         }
10    sc.close();
11 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Code

```
PS C:\Users\ym\.vscode> cd "c:\Users\ym\.vscode\r coding\" ; if ($?) { javac demo.java } ; if ($?) { java demo }
6
1 2 3 4 5 6
PS C:\Users\ym\.vscode\r coding> ^C
PS C:\Users\ym\.vscode\r coding> ; if ($?) { javac demo.java } ; if ($?) { java demo }
9
1 2 3 4 5 6 7 8 9
PS C:\Users\ym\.vscode\r coding> 
```


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J demo.java X

.vscode > r coding > J demo.java > ...

1import java.util.Scanner;

2public class demo {

Run | Debug

3public static void main(String[] args) {

4Scanner sc = new Scanner(System.in);

5int n = sc.nextInt();

6int sum = 0;

7for (int i = 1; i <= n; i++) {

8sum = sum + i;

9}

10System.out.println(sum);

11sc.close();

12}

13}

14

PROBLEMSOUTPUTDEBUG CONSOLETERMINALPORTS

Code + -

PS C:\Users\ym\.vscode> cd "c:\Users\ym\.vscode\r coding\" ; if (\$?) { javac demo.java } ; if (\$?) { java demo }

5

15

PS C:\Users\ym\.vscode\r coding> ^C

PS C:\Users\ym\.vscode\r coding> cd "c:\Users\ym\.vscode\r coding\" ; if (\$?) { javac demo.java } ; if (\$?) { java demo }

15

120

PS C:\Users\ym\.vscode\r coding>

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demo.java X

.vscode > r coding > J demo.java > demo

1 import java.util.Scanner;
2 public class demo{
3 public static void main(String[] args) {
4 Scanner sc = new Scanner(System.in);
5 int a = sc.nextInt();
6 int b = sc.nextInt();
7 int val = 1;
8 for (int i = 1; i <= b; i++) {
9 val = val * a;
10 }
11 System.out.println(val);
12 sc.close();
13 }
14 }

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\ym\.vscode> cd "c:\Users\ym\.vscode\r coding\" ; if (\$?) { javac demo.java } ; if (\$?) { java demo }
2
5
32
PS C:\Users\ym\.vscode\r coding> ^C
PS C:\Users\ym\.vscode\r coding> ; if (\$?) { javac demo.java } ; if (\$?) { java demo }
4
256
PS C:\Users\ym\.vscode\r coding>

Ln 14, Col 2 Spaces: 4 UTF-8 CRLF { } Ja